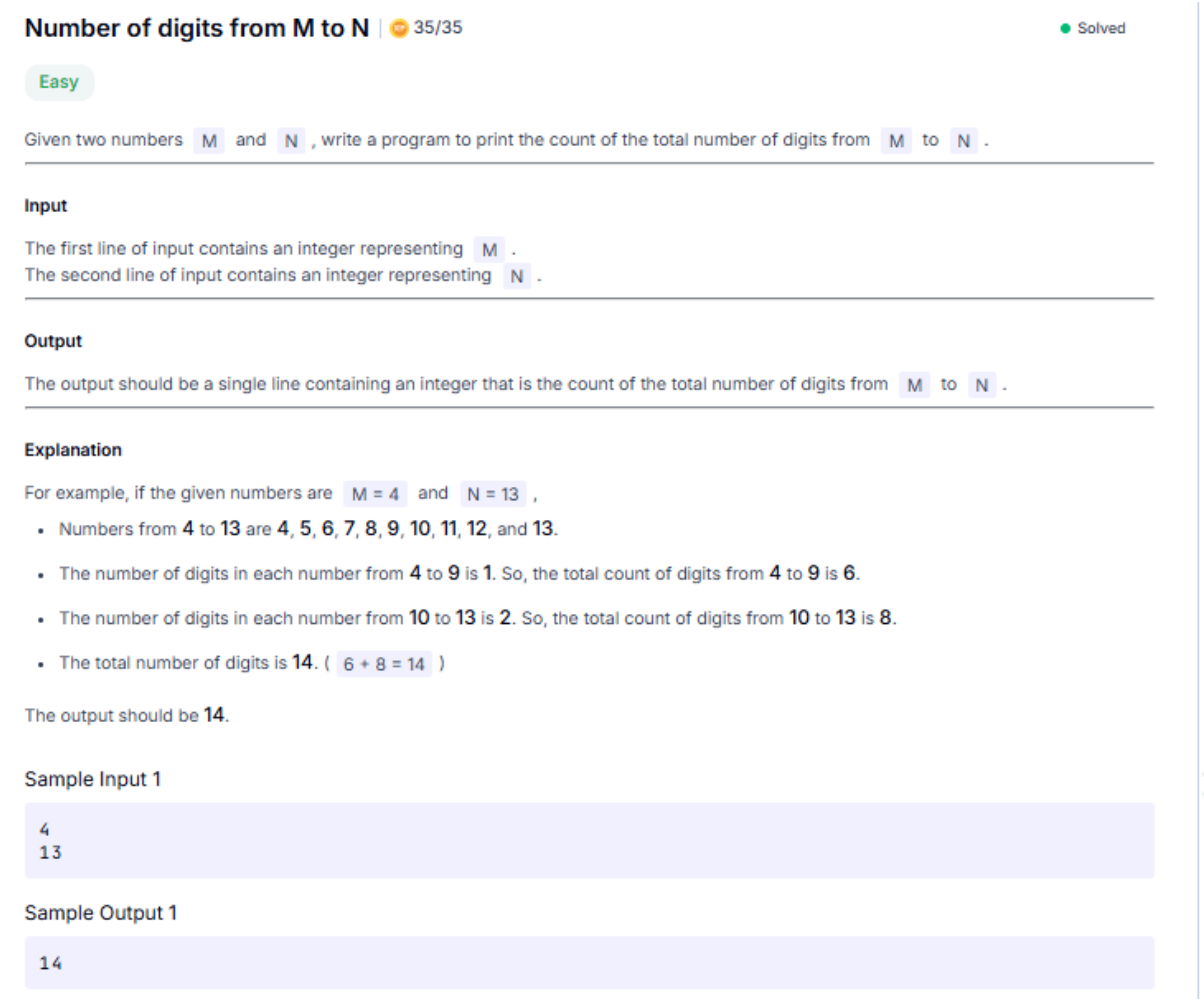
cp_10b

1



Number of digits from M to N | 35/35 Solved

Easy

Given two numbers `M` and `N`, write a program to print the count of the total number of digits from `M` to `N`.

Input

The first line of input contains an integer representing `M`.
The second line of input contains an integer representing `N`.

Output

The output should be a single line containing an integer that is the count of the total number of digits from `M` to `N`.

Explanation

For example, if the given numbers are `M = 4` and `N = 13`,

- Numbers from **4** to **13** are **4**, **5**, **6**, **7**, **8**, **9**, **10**, **11**, **12**, and **13**.
- The number of digits in each number from **4** to **9** is **1**. So, the total count of digits from **4** to **9** is **6**.
- The number of digits in each number from **10** to **13** is **2**. So, the total count of digits from **10** to **13** is **8**.
- The total number of digits is **14**. (`6 + 8 = 14`)

The output should be **14**.

Sample Input 1

```
4
13
```

Sample Output 1

```
14
```

code:

```
m=int(input())
n=int(input())
count=0
for i in range(m,n+1):
    i_len=len(str(i))
    count=count+i_len
print(count)
```

2

Divisible by 6 | 35/35

Solved

Easy

Given two numbers `M` and `N`, write a program to find the count of numbers from `M` to `N` that are divisible by **6**.

Print **No Numbers Found** if the count of numbers from `M` to `N` that are divisible by **6** is **0**. Otherwise, print the numbers from `M` to `N` that are divisible by **6** separated by a space.

Input

The first line of input contains an integer representing `M`.
The second line of input contains an integer representing `N`.

Output

The output should be a single line containing the string. **No Numbers Found** should be printed if the count of numbers from `M` to `N` that are divisible by **6** is **0**. Otherwise, space-separated numbers from `M` to `N` that are divisible by **6** should be printed.

Explanation

For example, if the given numbers are `M = 6` and `N = 23`,

- The numbers from **6** to **23** that are divisible by **6** are **6**, **12**, and **18**.
- So, the count of numbers from **6** to **23** that are divisible by **6** is **3**.
- The string obtained after joining the numbers from **6** to **23** that are divisible by **6** with a space is **6 12 18**.

The output should be **6 12 18**.

For example, if the given numbers are `M = 2` and `N = 5`,

- There are no numbers from **2** to **5** that are divisible by **6**.
- So, the count of numbers from **2** to **5** that are divisible by **6** is **0**.

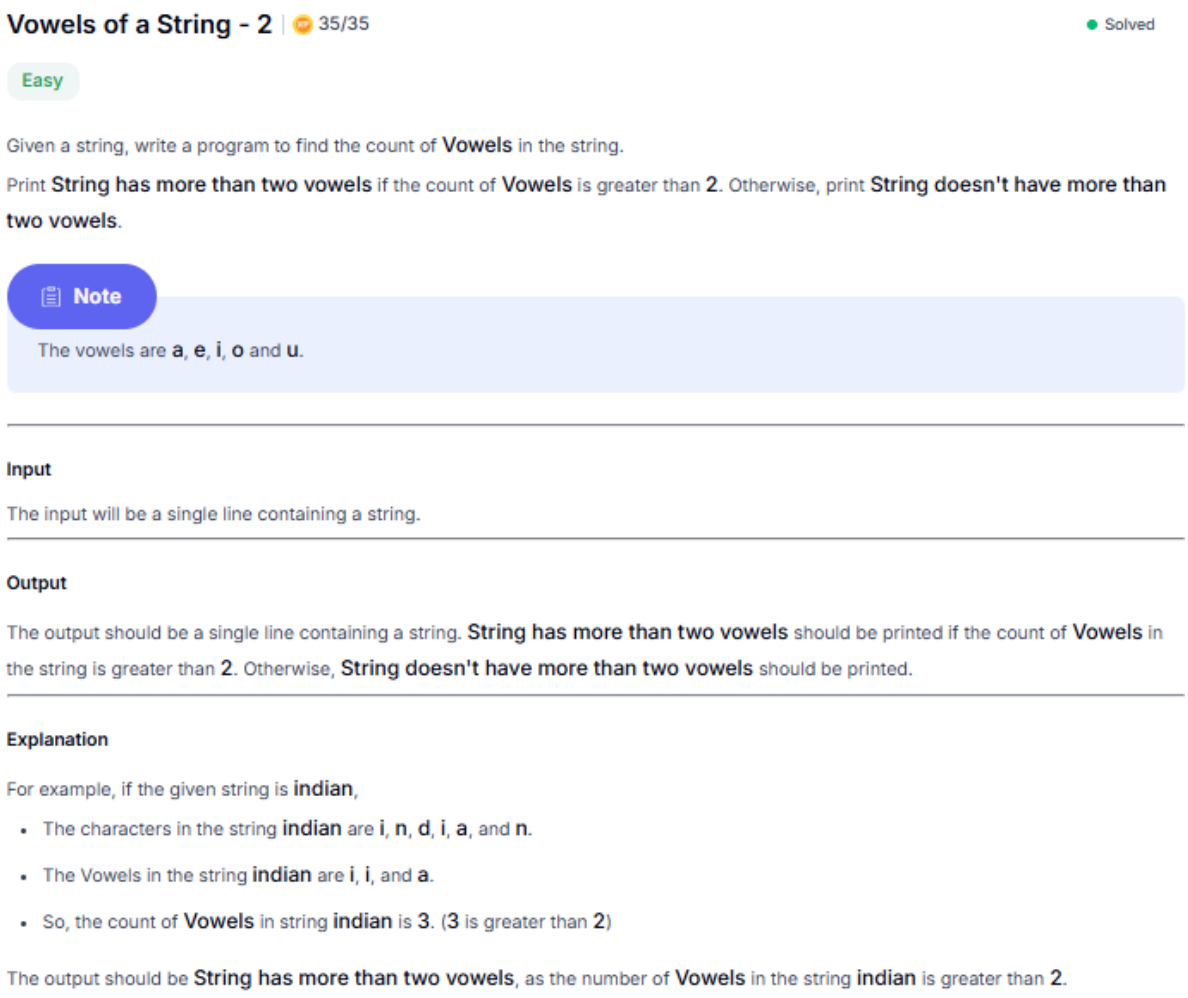
The output should be **No Numbers Found**.

code:

```
m=int(input())
n=int(input())
num_div_by_6=""
count=0
for i in range(m,n+1):
    if i%6==0:
        count=count+1
        num_div_by_6=num_div_by_6+str(i)+" "
if count==0:
    print("No Numbers Found")
```

```
else:
    print(num_div_by_6)
```

3



Vowels of a String - 2 | 35/35 ● Solved

Easy

Given a string, write a program to find the count of **Vowels** in the string.

Print **String has more than two vowels** if the count of **Vowels** is greater than **2**. Otherwise, print **String doesn't have more than two vowels**.

Note

The vowels are **a**, **e**, **i**, **o** and **u**.

Input

The input will be a single line containing a string.

Output

The output should be a single line containing a string. **String has more than two vowels** should be printed if the count of **Vowels** in the string is greater than **2**. Otherwise, **String doesn't have more than two vowels** should be printed.

Explanation

For example, if the given string is **indian**,

- The characters in the string **indian** are **i**, **n**, **d**, **i**, **a**, and **n**.
- The Vowels in the string **indian** are **i**, **i**, and **a**.
- So, the count of **Vowels** in string **indian** is **3**. (**3** is greater than **2**)

The output should be **String has more than two vowels**, as the number of **Vowels** in the string **indian** is greater than **2**.

code:

```
string=input()
count=0
for each_char in string:
    if each_char in ("a","e","i","o","u"):
        count=count+1
if count>2:
    print("String has more than two vowels")
```

```
else:
    print("String doesn't have more than two vowels")
```

4

Indivisible Number | 35/35 ● Solved

Easy

Given a number `N` , write a program that checks if `N` is **not** divisible by any number from **2** to **9**.

Print **Divisible Number** if the number `N` is divisible by any of the numbers from **2** to **9**. Print **Indivisible Number** if the number `N` is **not** divisible by any of the numbers from **2** to **9**.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing a string. **Divisible Number** should be printed if the number `N` is divisible by any of the numbers from **2** to **9**. **Indivisible Number** should be printed if the number `N` is **not** divisible by any of the numbers from **2** to **9**.

Explanation

For example, if the given number is `N = 15` ,

- **15** is divisible by **3** and **5**.
- **15** is **not** divisible by **2**, **4**, **6**, **7**, **8**, and **9**.

The output should be **Divisible Number**, as **15** is divisible by at least one of the numbers from **2** to **9**.

For example, if the given number is `N = 103` ,

- **103** is **not** divisible by any number from **2** to **9** (**2**, **3**, **4**, **5**, **6**, **7**, **8**, and **9**).

The output should be **Indivisible Number**, as **103** is **not** divisible by any number from **2** to **9**.

code:

```
num=int(input())
count=""
for i in range(2,10):
    divsible_2_n_9=(num%i)
    if (divsible_2_n_9==0):
        count="Divisible Number"
    elif count=="":
        count="Indivisible Number"
print(count)
```

5

Smallest Among N Numbers | 30/30

Solved

Write a program to print the smallest among the given N numbers.

Input

The first line of input will contain a positive integer (N).
The following N lines will contain an integer in each line.

Output

The output should be a single line containing the smallest among the given numbers.

Explanation

For example, if the given number is 5, then read the inputs in the next 5 lines and print the smallest among the given five numbers. If the given input integers in the next five lines are 8, 11, 96, 49, and 25, the output should be 8.

Sample Input 1

```
5
8
11
96
49
25
```

Sample Output 1

```
8
```

code:

```
n=int(input())
first_n=int(input())
smallest=first_n
for i in range(n-1):
    num=int(input())
    if num<smallest:
        smallest=num
print(smallest)
```

6

Divisible by 9 in a Range | 35/35

Solved

Easy

Given two numbers `M` and `N`, write a program to find the count of numbers from `M` to `N` that are divisible by **9**.

Print **No Numbers found** if the count of numbers from `M` to `N` that are divisible by **9** is **0**. Otherwise, print the numbers from `M` to `N` separated by a space that are divisible by **9**.

Input

The first line of input contains an integer representing `M`.
The second line of input contains an integer representing `N`.

Output

The output should be a single line containing the string. **No Numbers found** should be printed if the count of numbers from `M` to `N` that are divisible by **9** is **0**. Otherwise, space-separated numbers from `M` to `N` that are divisible by **9** should be printed.

Explanation

For example, if the given numbers are `M = 5` and `N = 25`,

- The numbers from **5** to **25** that are divisible by **9** are **9** and **18**.
- So, the count of numbers from **5** to **25** that are divisible by **9** is **2**.
- The string obtained after joining the numbers from **5** to **25** that are divisible by **9** with a space is **9 18**.

The output should be **9 18**.

For example, if the given numbers are `M = 3` and `N = 8`,

- There are no numbers from **3** to **8** that are divisible by **9**.
- So, the count of numbers from **3** to **8** that are divisible by **9** is **0**.

The output should be **No Numbers found**.

code:

```
m=int(input())
n=int(input())
count=0
num_div_by_9=""
for i in range(m,n+1):
    is_div_by_9=i%9
    if is_div_by_9==0:
       count=count+1
       num_div_by_9=num_div_by_9+str(i)+" "
if count==0:
    print("No Numbers found")
else:
    print(num_div_by_9)
```

7

2 Series | 35/35

Solved

Easy

Given a number `N` , write a program to print the `N` terms in the given series, each on a new line.

Series:

```
2, 22, 222, 2222, .... N terms
```

Terms in the given series:

Term Number	Term	Number of 2's
First Term	2	1
Second Term	22	2
Third Term	222	3
..	..	..
Nth Term	2222... (upto N)	N

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` lines containing the terms from **First Term** to **Nth Term** in the given series.

Explanation

For example, if the given number is `N = 6` ,

- The number of terms is **6**.
- Print the terms from **First Term** to **Sixth Term** in the given series.

code:

```python
n=int(input())
for i in range(1,n+1):
    row=str(2)*i
    print(row)
```

8

Sum of N Terms in 2 Series | 40/40

Solved

Given an integer `N` , write a program to print the **sum** of `N` terms in the given series.

Series:

```
2, 22, 222, ... N terms
```

Terms in the given series:

Term Number	Term	Number of 2's
First Term	2	1
Second Term	22	2
Third Term	222	3
..	..	..
Nth Term	222... (up to N)	N

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing an integer that is the **sum** of the series from **First Term** to **Nth Term**.

Explanation

For example, if the given number is `N = 3` ,

- The number of terms is **3**.
- The terms from **First Term** to **Third Term** in the given series are **2**, **22** and **222**.

Terms in the given series:

Term Number	Term	Number of 2's
First Term	2	1
Second Term	22	2
Third Term	222	3

- The **sum** of the terms from **First Term** to **Third Term** is **246**. (`2 + 22 + 222 = 246`)

The output should be **246**.

code:

```
n=int(input())
sum=0
for i in range(1,n+1):
    row=str(2)*i
    int_row=int(row)
```

```
    sum=sum+int_row
print(sum)
```

9

Sum of N Terms 1 series | 30/30

Solved

Easy

Given an integer `N` , write a program to print the **sum** of `N` terms in the given series.

Series:

```
1, 11, 111, ... N terms
```

Terms in the given series:

Term Number	Term	Number of 1's
First Term	1	1
Second Term	11	2
Third Term	111	3
..	..	..
Nth Term	111... (up to N)	N

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing an integer that is the **sum** of the series from **First Term** to **Nth Term**.

Explanation

For example, if the given number is `N = 4` ,

- The number of terms is **4**.
- The terms from **First Term** to **Fourth Term** in the given series are **1**, **11**, **111**, and **1111**.

Terms in the given series:

Term Number	Term	Number of 1's
First Term	1	1
Second Term	11	2
Third Term	111	3
Fourth Term	1111	4

- The **sum** of the terms from **First Term** to **Fourth Term** is **1234**. (`1 + 11 + 111 + 1111 = 1234`)

The output should be **1234**.

code:

```
n=int(input())
sum=0
for i in range(1,n+1):
    row=str(1)*i
    int_row=int(row)
    sum=sum+int_row
print(sum)
```

10

Sum of N Terms in X Square Series | 35/35

Solved

Easy

Given two numbers `X` and `N` , write a program to print the **sum** of `N` terms in the given series.

Series:

```
(X)², (XX)², (XXX)², ... N terms
```

Terms in the given series:

Term Number	Term	Number of X's
First Term	$(X)^2$	1
Second Term	$(XX)^2$	2
Third Term	$(XXX)^2$	3
..	..	..
Nth Term	(XXXX...upto N)^2	N

Input

The first line of input contains an integer representing `X` .
The second line of input contains an integer representing `N` .

Output

The output should be a single line containing an integer that is the **sum** of the series from the **First Term** to the **Nth Term**.

Explanation

For example, if the given numbers are `X = 4` and `N = 3` ,

- The number of terms is **3**.
- The terms from **First term** to **Third term** in the given series are **16** (4^2), **1936** (44^2), and **197136** (444^2).

  Terms in the given series:

Term Number	Term	Number of X's (4's)
First Term	4^2 (16)	1
Second Term	44^2 (1936)	2
Third Term	444^2 (197136)	3

- The **sum** of the terms from the **First Term** to the **Third Term** is **199088**. (`16 + 1936 + 197136 = 199088`)

The output should be **199088**.

code:

```
m=int(input())
n=int(input())
sum=0
for i in range(1,n+1):
```

```
    row=str(m)*i
    sq=int(row)**2
    sum=sum+sq
print(sum)
```